Daniel Ansah Akrofi

Data Science and Insights

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Portfolio | GitHub | LinkedIn | Warwick profile

Profile

Warwick Business School PhD researcher with a background in commercial strategy and consumer research. I use statistical modelling and natural language processing (NLP) to study product and market behaviour and communicate findings through charts, briefings and recommendations.

Technical Skills

Programming and data: Python (pandas, NumPy, scikit-learn, Matplotlib), SQL (PostgreSQL), R (RStudio), web scraping, data cleaning, Git/GitHub

Statistics and causal inference: Regression, Bayesian time-series modelling, difference-in-differences (DiD), event studies, cross-validation

Machine learning and NLP: Feature engineering, text classification, BERTopic, sentiment analysis

Dashboards and reporting: Microsoft Power BI, Microsoft Excel dashboards, KPI reporting, data visualisation, data storytelling

Doctoral Research

PhD Researcher in Business and Management

Warwick Business School, Coventry | Oct 2024 - Present

- Built a Python pipeline to align Steam discussion, sales estimates, search trends and news coverage to release dates for 1,695 games. The discussion corpus contained 7.96 million posts and comments.
- Fine-tuned DistilRoBERTa models to detect speculation and uncertainty cues. Used BERTopic to group posts by product topic and validated the pipeline against 500 independently annotated sentences.
- Used ordinary least squares regression to study sales, revenue and reviews. Holding out entire publishers in cross-validation showed that speculation measures improved sales and revenue predictions beyond standard buzz metrics.
- Developing a local-projections difference-in-differences event study and Bayesian structural time-series models to estimate how pre-release announcements shift uncertainty in Steam discussions, using matched games at the same stage before launch.

Selected Projects

Online Community Launch Analytics | MRes Thesis | GitHub

- Used Python, BERTopic and VADER to identify product complaints on Reddit and score replies. Used regression and charts to show that more hopeful language before a complaint was associated with a higher share of positive replies.
- Published a reproducible sample workflow using pandas to prepare the data, scikit-learn to fit a baseline model and Matplotlib to chart the results.

Movie League | Film Performance Tracker | GitHub

- Scraped box-office and review data with Python and Beautiful Soup. Queried and updated PostgreSQL tables with SQL, combining box-office growth, audience grades and critic scores into a daily ranking.

Anime League | R Modelling Project

- Used R in RStudio to explore which title characteristics were associated with audience ratings. Encoded title type and fitted a multiple linear regression and an exploratory regression tree with rpart.

Additional Experience

Postgraduate Teaching Assistant

Warwick Business School, Coventry | Nov 2024 - Jul 2026

- Led postgraduate seminars in Marketing Strategy and Digital Marketing Technology and Management, using industry cases to discuss how platform data informs marketing decisions.
- Received Warwick Business School's Outstanding Contribution to Teaching award in 2024/25 and 2025/26.

Postgraduate Research Assistant

Warwick Business School, ISMA Group, Coventry | Jan 2024 - Jun 2024

- Built a workflow to track developments in AI adoption and digital platforms. Synthesised academic research and industry evidence into weekly slide decks for academics and practitioners.

Communication Strategy Consultant

Independent, Accra | Sep 2020 - Aug 2022

- Turned audience research into campaign messaging and stakeholder briefings for clients including Standard Chartered and Ghana's Ministry of Finance.

Strategy and Creative Lead

Adams 360 Marketing Agency, Accra | May 2016 - May 2019

- Led consumer and market research across 10 international brands. Presented findings and recommendations to clients to guide positioning and campaign decisions.
- Translated consumer and competitor research into launch recommendations for Total Excellium in Ghana. The client recorded 11.9% year-on-year revenue growth in 2016-17.

Strategic Planner

Adams 360 Marketing Agency, Accra | May 2013 - Apr 2016

- Used customer and market insights to plan digital campaigns for two FMCG brands. Each gained more than 20,000 social media followers and increased online impressions by over 500%.

Education

PhD in Business and Management

Warwick Business School | 2024 - Present

MRes in Business and Management

Warwick Business School | 2024 | Distinction

MSc in Marketing and Strategy

Warwick Business School | 2020 | Distinction. Thesis: statistical analysis of how key personnel were associated with film box-office performance.

BSc in Business Administration

Ashesi University | 2013 | Cum Laude